



Bahir Dar University
**Faculty of computing department of software
engineering**

Operating system and system programming
Individual Assignment

Installation of chrome os

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1.Introduction

Background

Chrome OS is based on the Linux kernel and is built to support fast and efficient computing with minimal system requirements. It mainly uses the Chrome browser as its core interface, where most applications run online. The system is designed with strong security features such as sandboxing, verified boot, and automatic updates, which help protect user data and improve system reliability. As cloud computing becomes more common, Chrome OS represents a new approach to operating systems that depend more on internet services than traditional software installed on a computer.

Motivation

The motivation is to gain a deeper understanding of how modern operating systems are evolving toward cloud-based and lightweight architectures, using **Chrome OS** as a practical example. Unlike traditional systems that rely on locally installed software, Chrome OS emphasizes web applications, online storage, and continuous connectivity. Studying this approach helps in understanding current trends in computing, especially the shift toward cloud-centered environments. It also provides insight into how such systems achieve speed, efficiency, and security with minimal hardware resources.

Another important motivation is to explore virtualization technology using tools such as VMware Workstation or Oracle VM VirtualBox. Virtualization allows Chrome OS to run in an isolated and controlled environment without affecting the main operating system. This makes it possible to safely perform installation, configuration, and performance analysis while developing practical technical skills. Additionally, it helps in evaluating the advantages and limitations of Chrome OS in real-world scenarios, providing a better understanding of its usability and effectiveness.

2. OBJECTIVES

The primary objectives of this project-work documentation are as follows:

- ✓ To successfully deploy ChromeOS (specifically the Flex version or ChromiumOS equivalent) within a Virtual Environment Tool (OracleVM VirtualBox).
- ✓ To analyze the underlying architectural differences between ChromeOS and traditional monolithic kernel operating systems.
- ✓ To document the installation lifecycle, identifying potential hardware/software bottlenecks during virtualization.
- ✓ To evaluate the filesystem compatibility and performance characteristics of the OS.
- ✓ To provide a comprehensive guide for troubleshooting common virtualization issues associated with ChromeOS.

3. REQUIREMENTS

To successfully install and run Chrome OS in a virtual environment, certain hardware, software, and configuration requirements must be fulfilled. These requirements ensure that the system runs smoothly, efficiently, and without performance issues during virtualization.

3.1 Hardware Requirements

Hardware plays a very important role in virtualization because the virtual machine shares resources with the host operating system. Insufficient hardware may result in slow performance, lagging, or installation failure.

- **Processor (CPU):** Intel Core i3/i5/i7 (8th Gen or newer) or AMD Ryzen equivalent with VT-x/AMD-V virtualization support enabled in BIOS.
- **Memory (RAM):** Minimum 8GB (4GB dedicated to the Virtual Machine).
- **Storage:** 20GB of available Solid State Drive (SSD) space for the VM disk image.
- **Graphics (GPU):** Integrated or discrete GPU with support for DirectX 11 or OpenGL 4.x for UI acceleration.

3.2 SOFTWARE REQUIREMENTS

I. Host Operating System:

The host operating system is the main operating system already installed on the computer that supports virtualization software. **It controls the computer hardware and provides the environment where virtual machines run.**

Examples:

- Windows 10 / Windows 11
- Linux (Ubuntu, Fedora, Debian)

II. Virtualization Software (Hypervisor)

Virtualization software is a program that creates and manages virtual machines, allowing multiple operating systems to run on one computer. It acts like a “virtual computer creator” that allows Chrome OS to run inside your system without affecting the real operating system.

Examples:

- Oracle VM VirtualBox
- VMware Workstation Player

III. Chrome OS Image File

A Chrome OS image file is a pre-built operating system file used to install or boot Chrome OS in a virtual machine. It contains all system files required to run Chrome OS.

Examples:

- Chromium OS image
- CloudReady image
- File formats: ISO, VMDK

IV. Virtualization Support Tools

These are additional software tools installed inside the virtual machine to improve performance and integration between host and guest systems. They improve mouse control, screen resolution, graphics performance, and file sharing between systems.

Examples:

- VirtualBox Guest Additions
- VMware Tools

V. BIOS/UEFI Firmware Settings

BIOS/UEFI is low-level system software that controls hardware before the operating system starts. It must be enabled to allow virtualization software to run properly.

Important Feature: Virtualization Technology (VT-x / AMD-V)

VI. Device Drivers (Optional but Important)

Definition:

Drivers are software that allow the operating system to communicate with hardware devices. They help improve performance and ensure smooth operation inside the virtual machine.

Examples:

- Graphics drivers
- Network drivers

4. Installation Steps

Step 1: Install Virtualization Software(OracleVM VirtualBox).

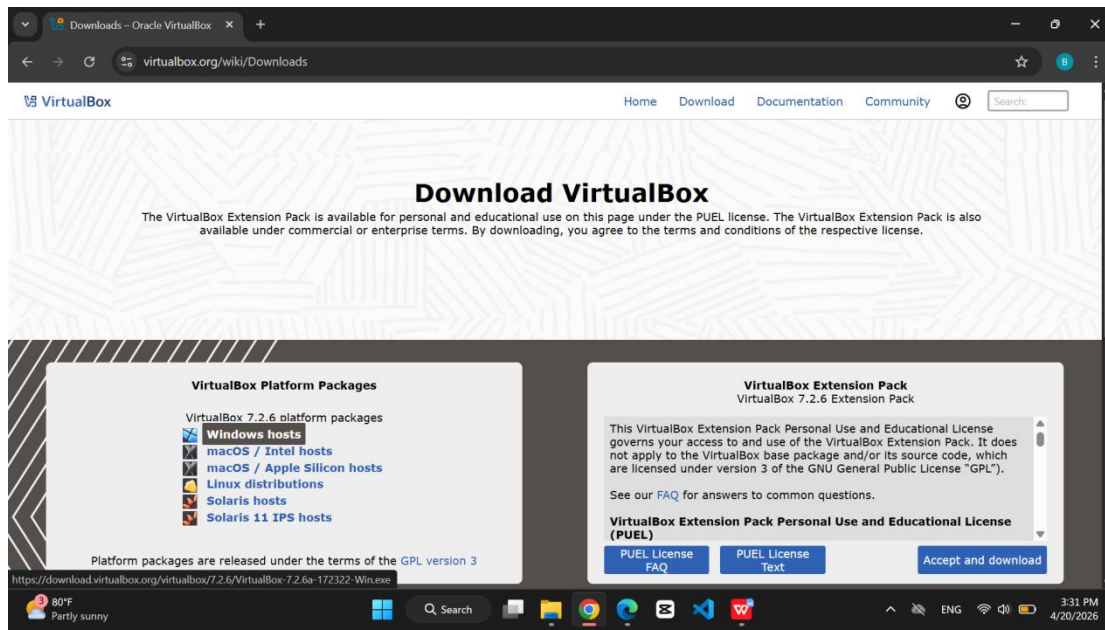


Figure 1.1: Downloading the Oracle VM VirtualBox installer, ensuring the version matches the host architecture (Windows x64).

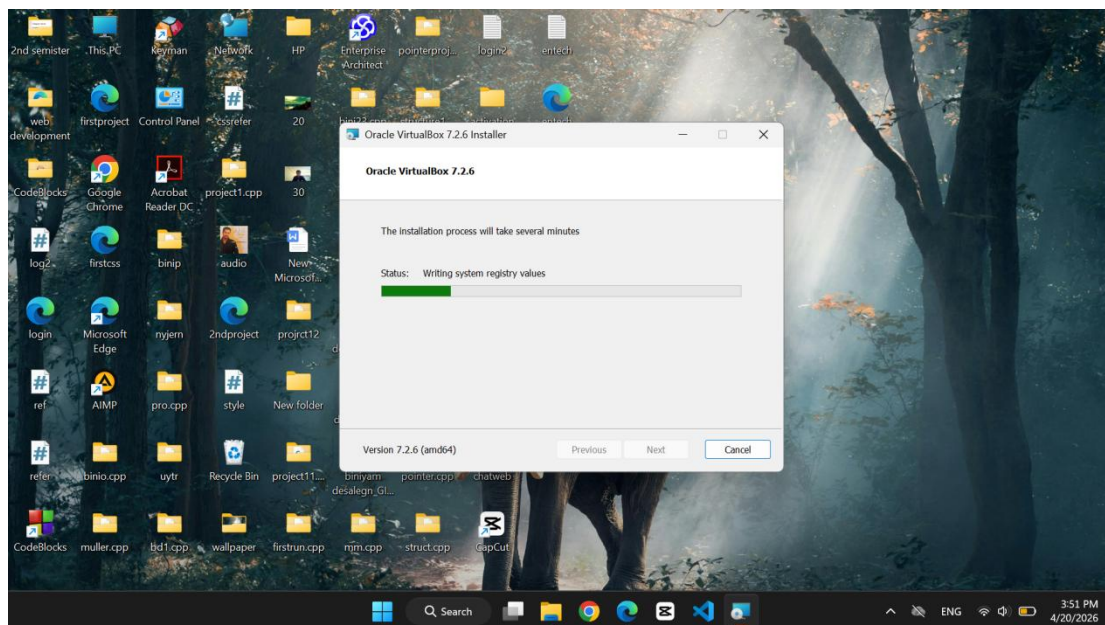


Figure 1.2: Executing the installation wizard to configure virtual network kdrivers and system registry entries.

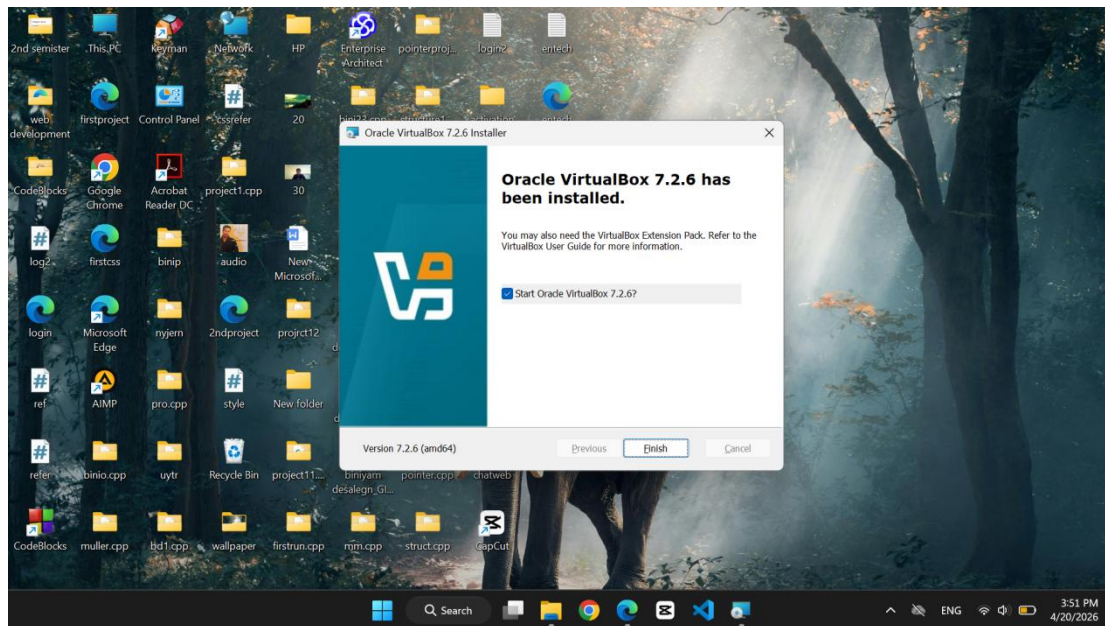


Figure 1.3: Successful completion of the hypervisor setup, providing the platform required to create and manage the ChromeOS virtual machine.

Step 2: Download Chrome OS Flex and Convert Image.

The ChromeOS Flex image was downloaded directly from Google's official server as a compressed .bin file, which was later extracted and used for virtual machine setup.

1. Download Chrome OS Flex

Chrome OS Flex is downloaded from the official Google website using the **Chromebook Recovery Utility**. Instead of an ISO file, it downloads a recovery image.

2. Convert .BIN to .VDI using CMD(Command Prompt)

To use the image in Oracle VirtualBox, the downloaded .bin file is converted into .vdi format using Command Prompt. This is necessary because VirtualBox cannot directly support .bin files.

Steps:

1. Open Command Prompt (Administrator Mode)

1. Search for **CMD** in the Start menu.
2. Right-click on **Command Prompt**.
3. Select **Run as Administrator**.

This step allows system-level commands to be executed.

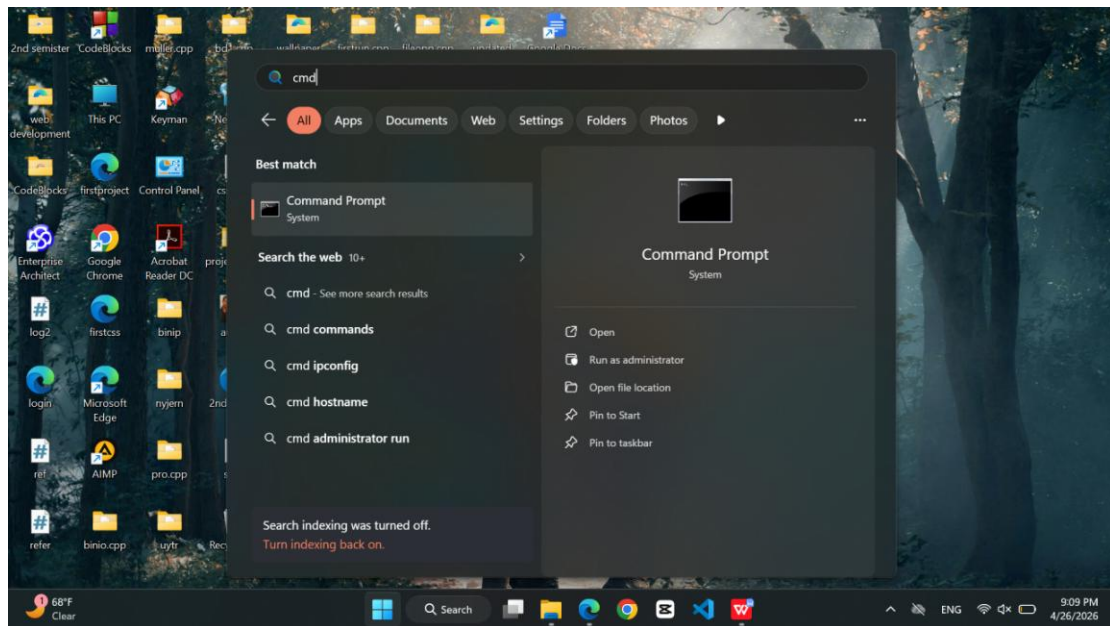


Figure 2.1: Searching for Command Prompt (CMD) in the Start menu to run as administrator.

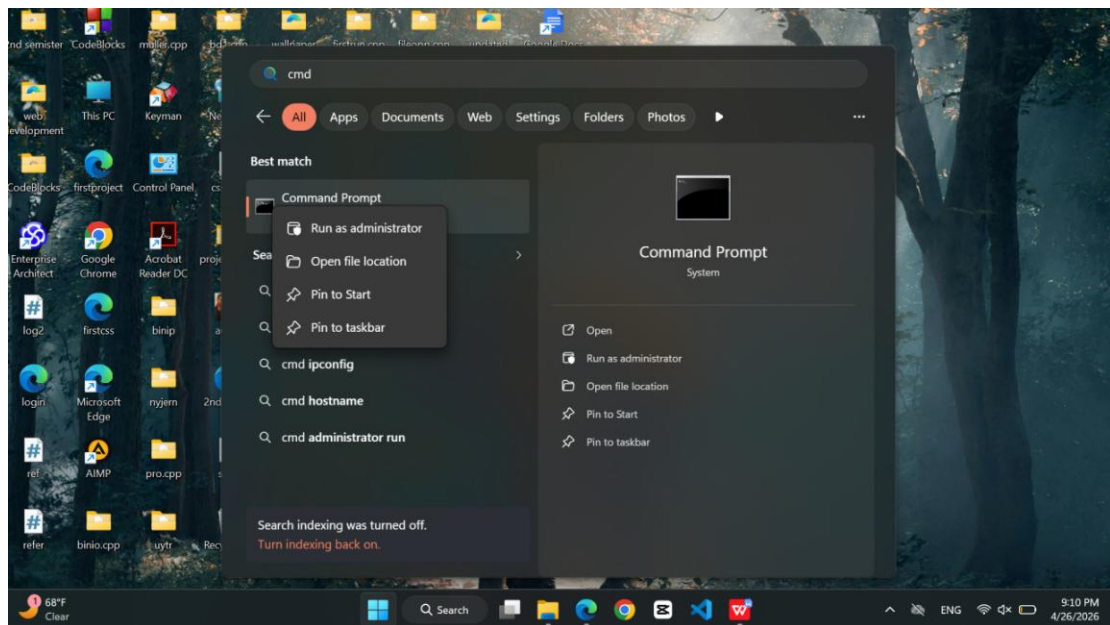


Figure 2.2: Right-clicking Command Prompt to select "Run as administrator" for system-level execution.

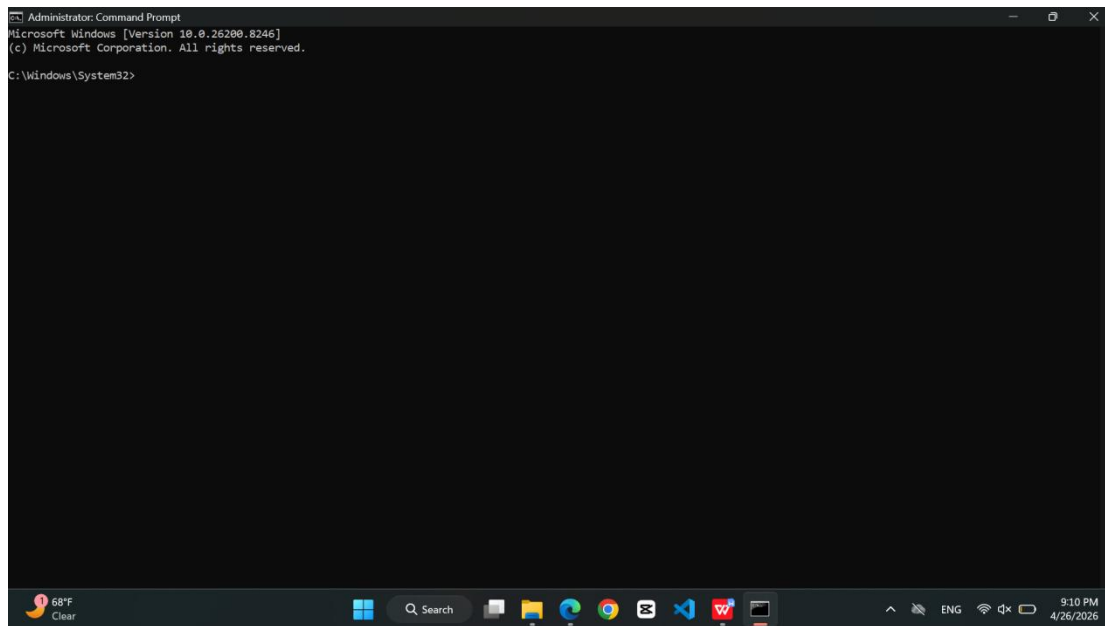


Figure 2.3: Administrator Command Prompt window ready for command execution.

2. Go to VirtualBox folder:

Run the following command:

```
cd "C:\Program Files\Oracle\VirtualBox"
```

This command is used to navigate to the installation directory of Oracle VirtualBox.

3. Run conversion command(Convert .BIN File to .VDI Format)

Execute the following command:

```
VBoxManage convertfromraw "C:\Users\HP\Downloads\chromeos_flex.bin"  
"C:\Users\HP\Downloads\chromeos.vdi" --format VDI
```

What this command does:

- Takes the downloaded **ChromeOS Flex .bin file** (input)
- Converts it into **.vdi format** (output)
- Saves it in the Downloads folder

4. Wait until conversion completes

After running the command, the system starts processing the file. It reads the entire image and converts it into a VirtualBox-compatible format.

Output file: **chromeos.vdi**

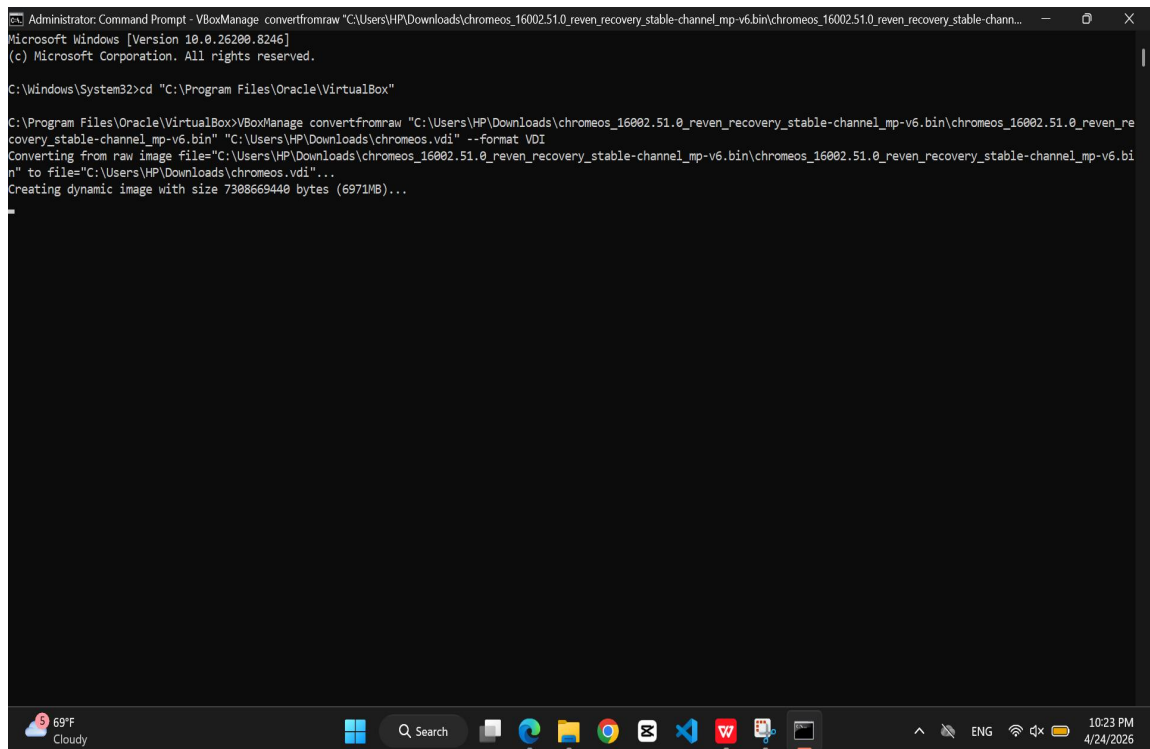


Figure 2.4: Conversion of ChromeOS Flex .bin file to .vdi format using Command Prompt (VBoxManage tool)

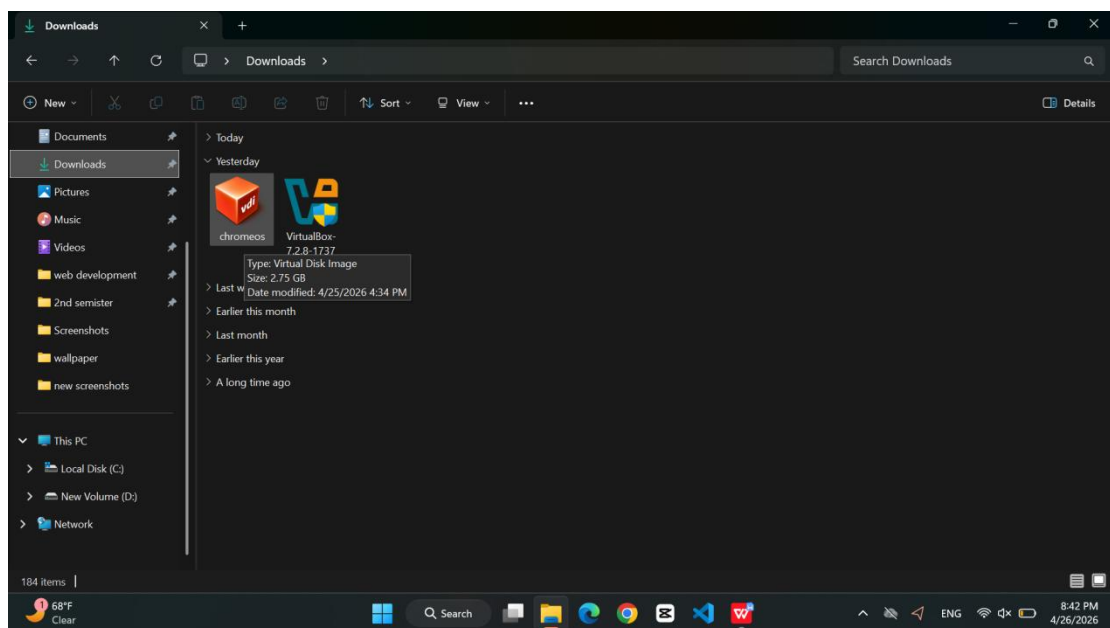


Figure 2.5: converted chrome os .vdi file

Step 3: Creating Virtual Machine in Oracle VirtualBox

After successfully converting the Chrome OS Flex image from **.bin** to **.vdi** format, the next step is to create a virtual machine using Oracle VirtualBox.

Step 3.1 : Create a New Virtual Machine

1. Open Oracle VirtualBox.
2. Click on **New**.
3. Enter the following details:
 1. **Name:** Chrome os
 2. **Type:** Linux
 3. **Version:** Other Linux (64-bit)
4. Select **Skip Unattended Installation**.
5. Leave the ISO image field empty.
6. Click **Next**.

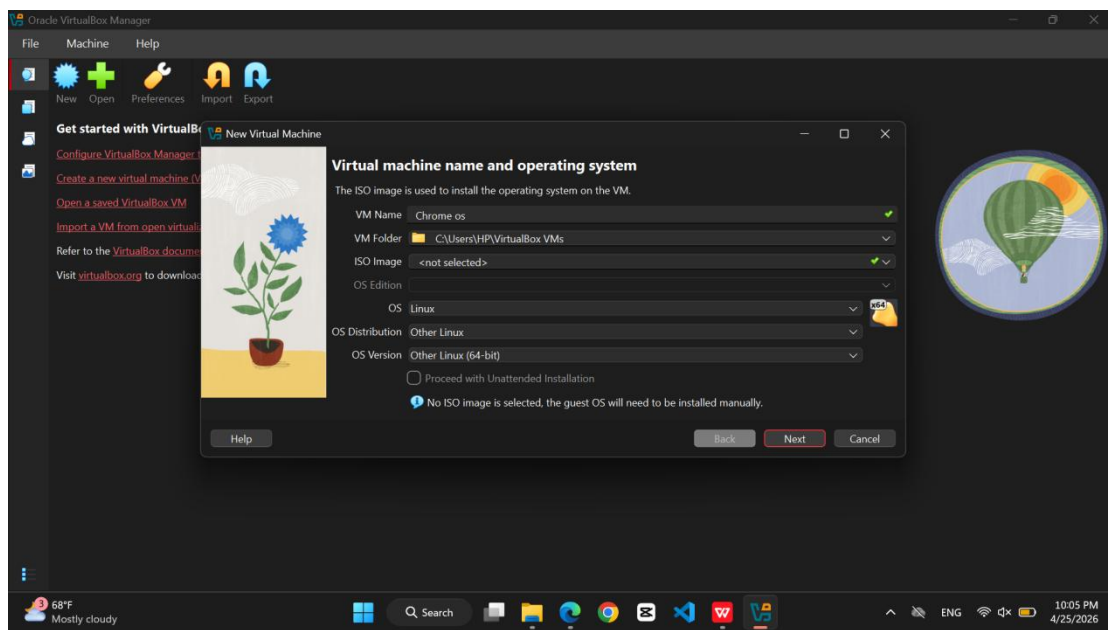


Figure 3.1: Virtual machine name and operating system configuration screen in Oracle VirtualBox.

Step 3.2: Allocate Hardware Resources

1. Set the **Base Memory (RAM)** to **4096 MB (4 GB)**.
2. Set the **Number of CPUs** to **2 processors**.
3. Click **Next**.

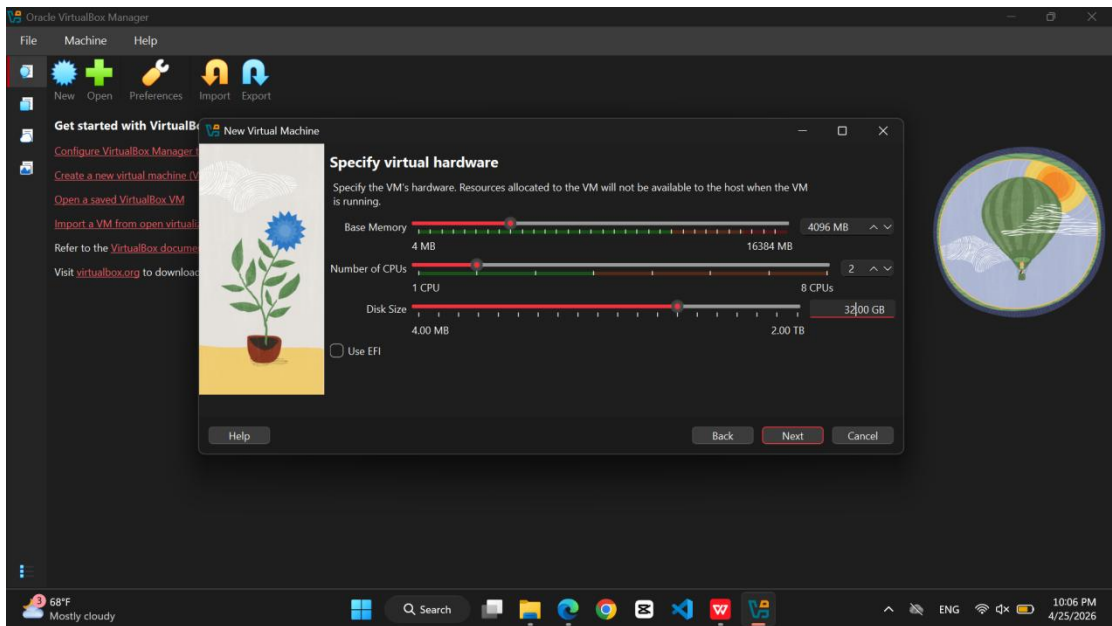


Figure 3.2: Specifying virtual hardware resources (RAM and CPU) for the new virtual machine.

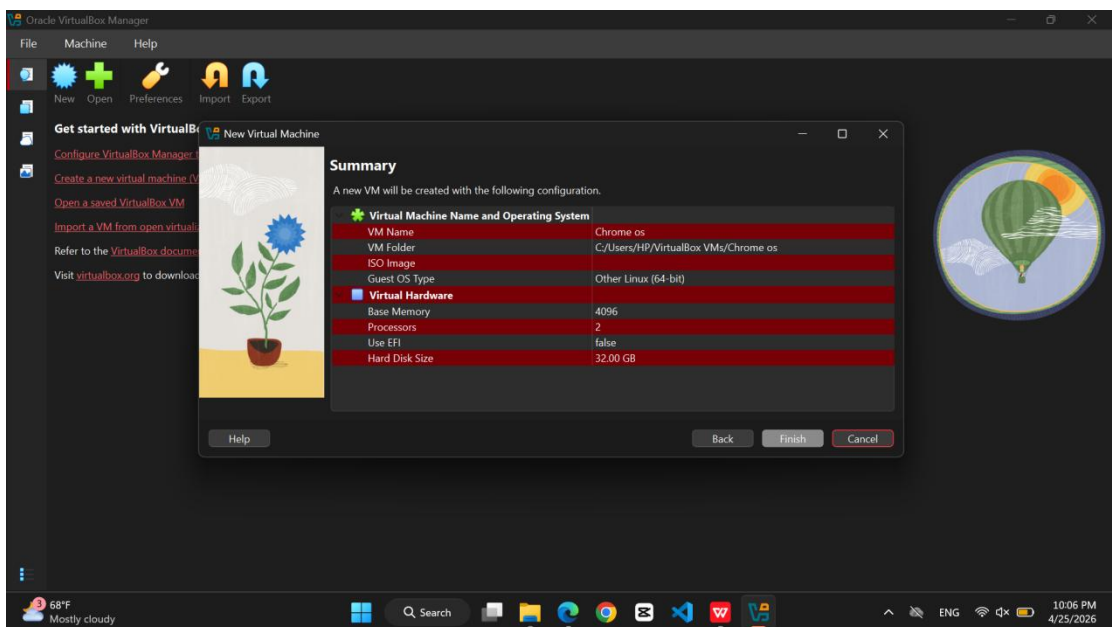


Figure 3.3: Summary of the virtual machine configuration before final creation.

Step 4: Configure Virtual Machine Settings

After creating the virtual machine, additional configuration is required.

Step 4.1: System Configuration

1. Go to **Settings** → **System** → **Motherboard**.
2. Set Boot Order:
 1. Hard Disk (First)

2. Disable Floppy and Optical
3. Enable **EFI** (Enable EFI for special OSes).

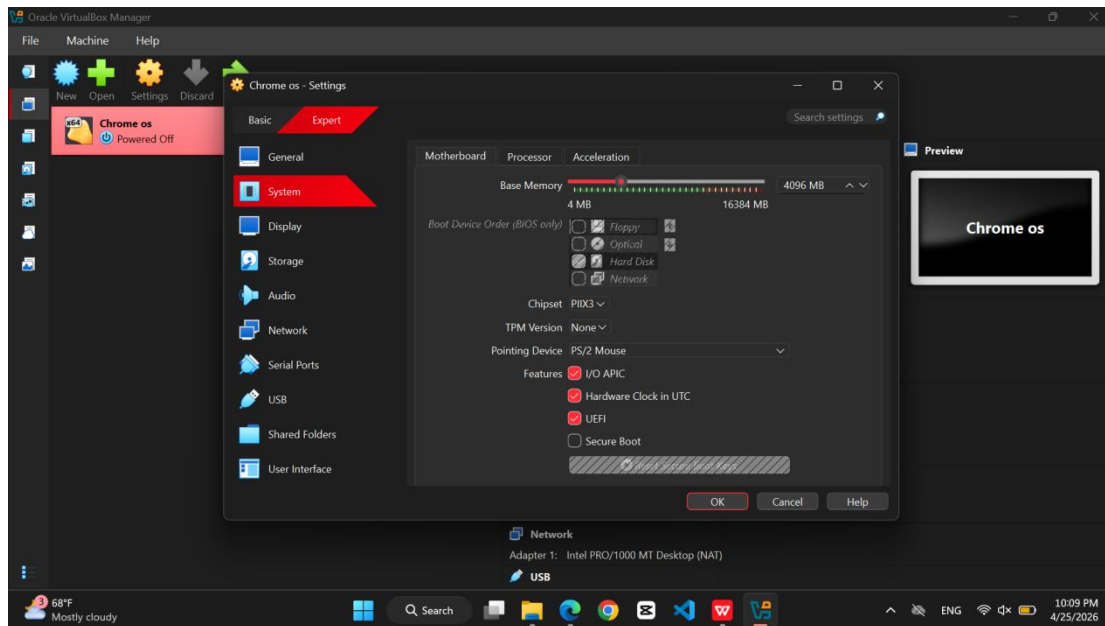


Figure 4.1: System motherboard settings including boot order and EFI enablement.

Step 4.2: Processor Configuration

1. Go to **Settings** → **System** → **Processor**.
2. Set CPU to **2 processors**.
3. Enable **PAE/NX**.

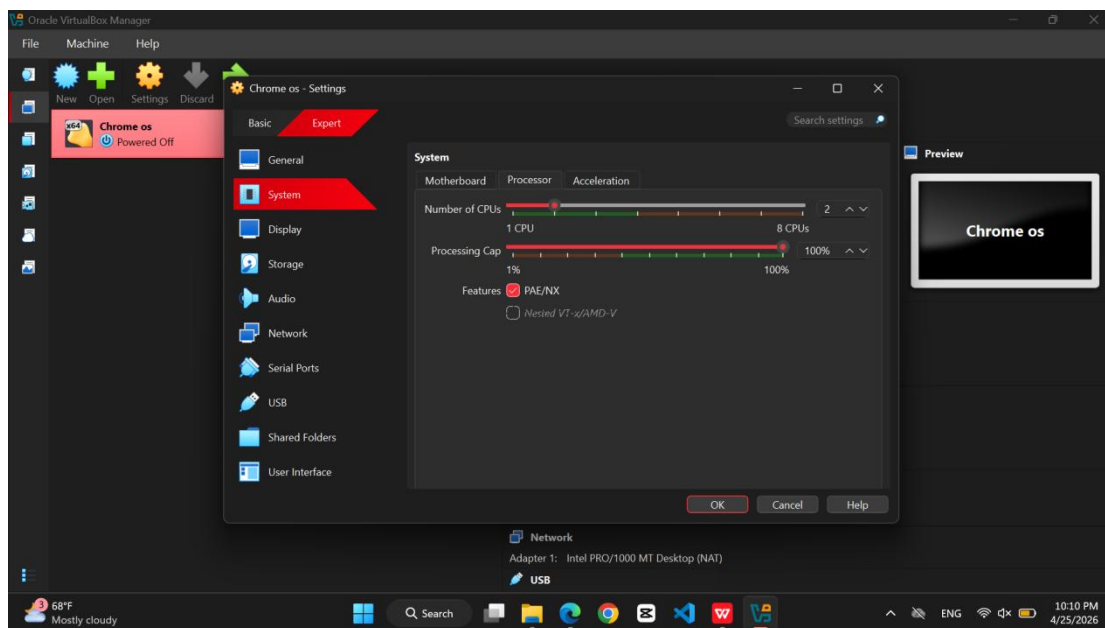


Figure 4.2: Processor configuration settings showing CPU allocation and PAE/NX enablement.

Step 4.3: Display Configuration

1. Go to **Settings** → **Display**.
2. Set Video Memory to **128 MB**.
3. Enable **3D Acceleration**.

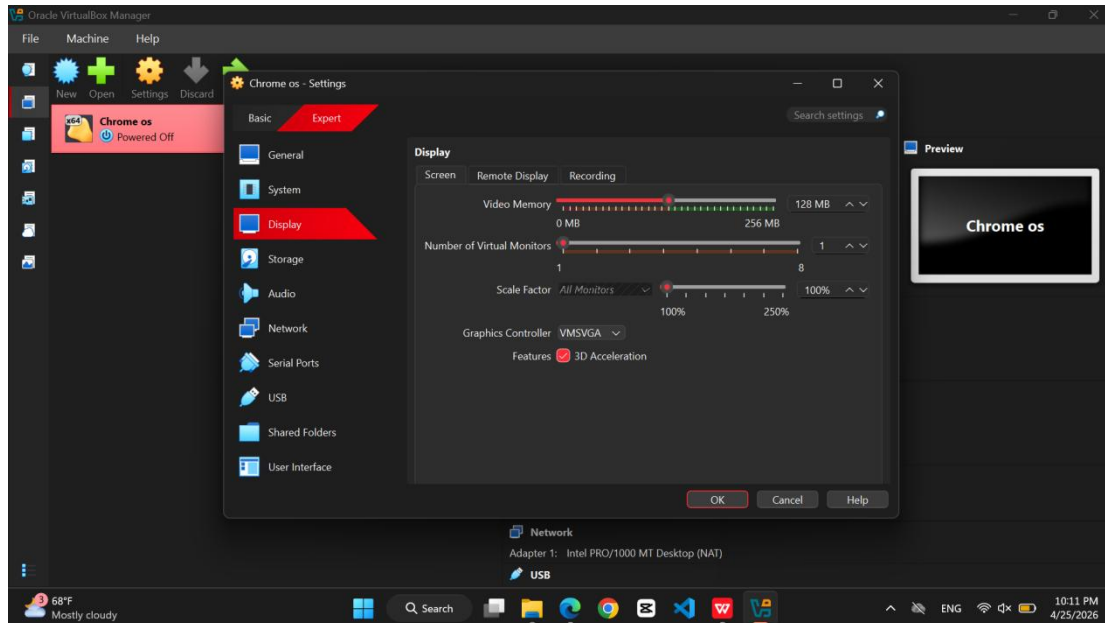


Figure 4.3: Display configuration settings for video memory and 3D acceleration.

Step 4.4: Storage Verification

1. Go to **Settings** → **Storage**.
2. Ensure the following:
 1. The file **chromeos.vdi** is attached under SATA Controller.
 2. The Optical Drive is empty (no ISO file attached).

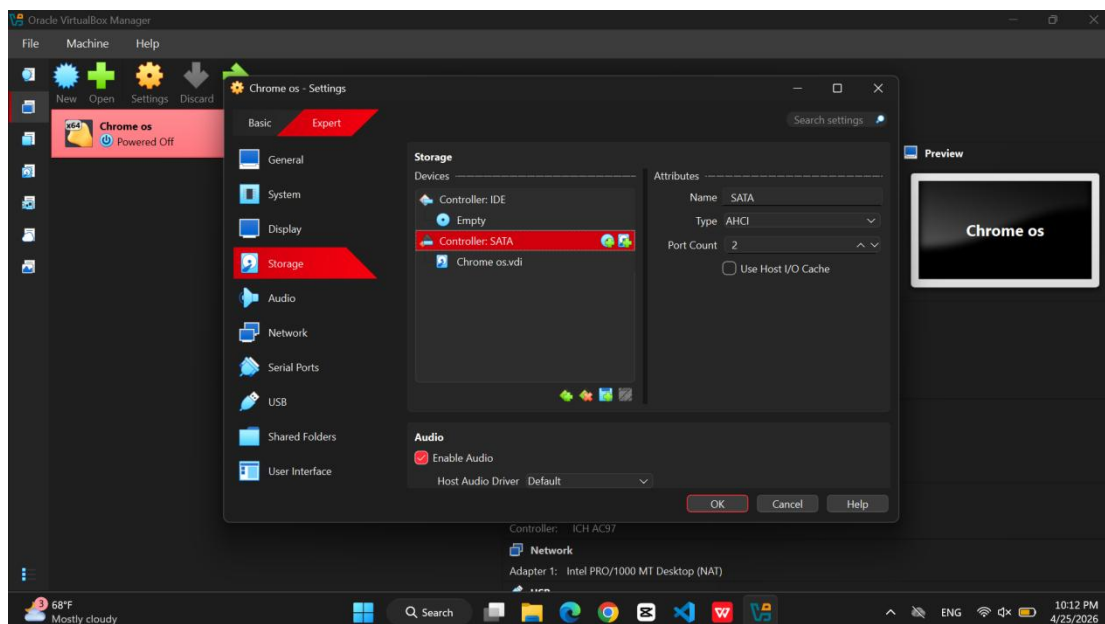


Figure 4.4: Storage verification showing the attached .vdi file under the SATA controller.

Step 5: Start the Virtual Machine

1. Click on the virtual machine.
2. Click the **Start** button.
3. The system will attempt to boot Chrome OS Flex.

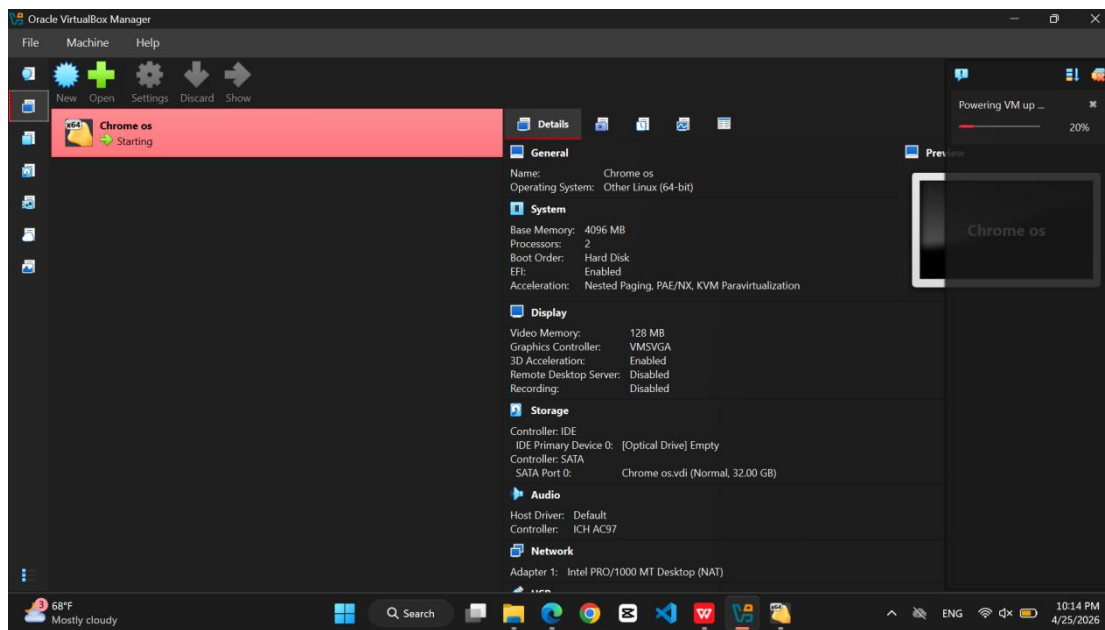


Figure 5: Starting the virtual machine to begin the ChromeOS Flex boot process.

Step 6: Boot Screen

1. After starting, the system begins loading the operating system.
2. ChromeOS Flex boot process starts.

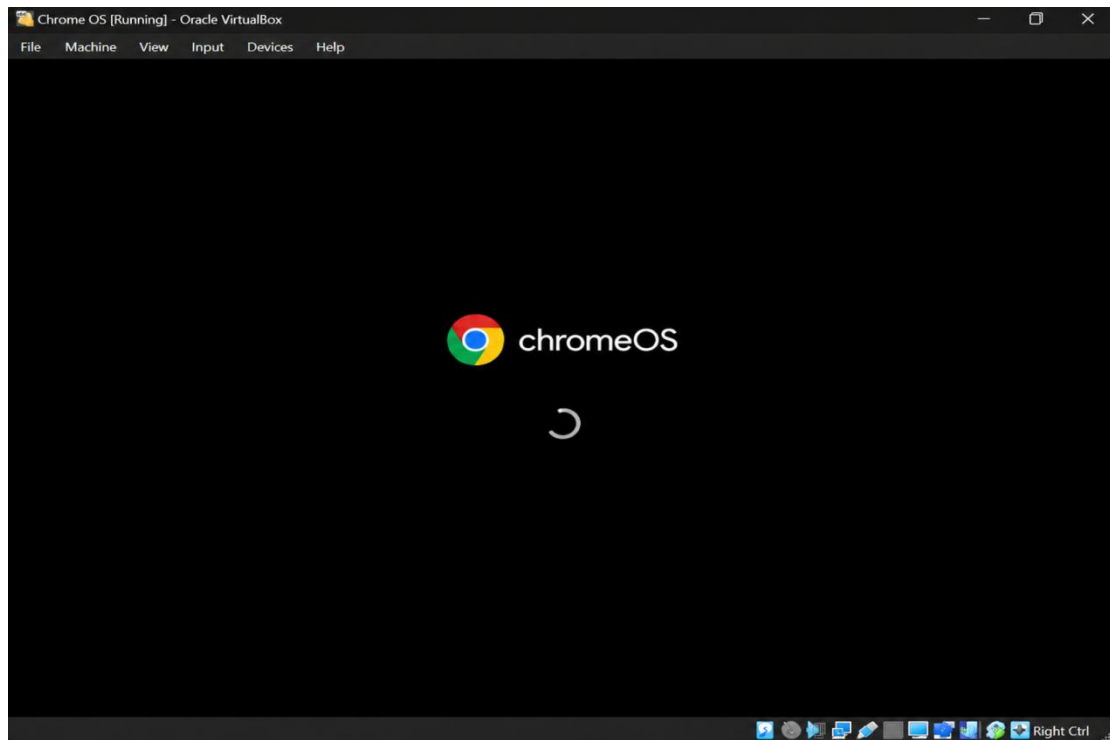


Figure 6: ChromeOS Flex boot screen showing the operating system loading.

Step 7: Welcome Screen

1. The system loads the first setup screen.
2. User is prompted to begin setup.

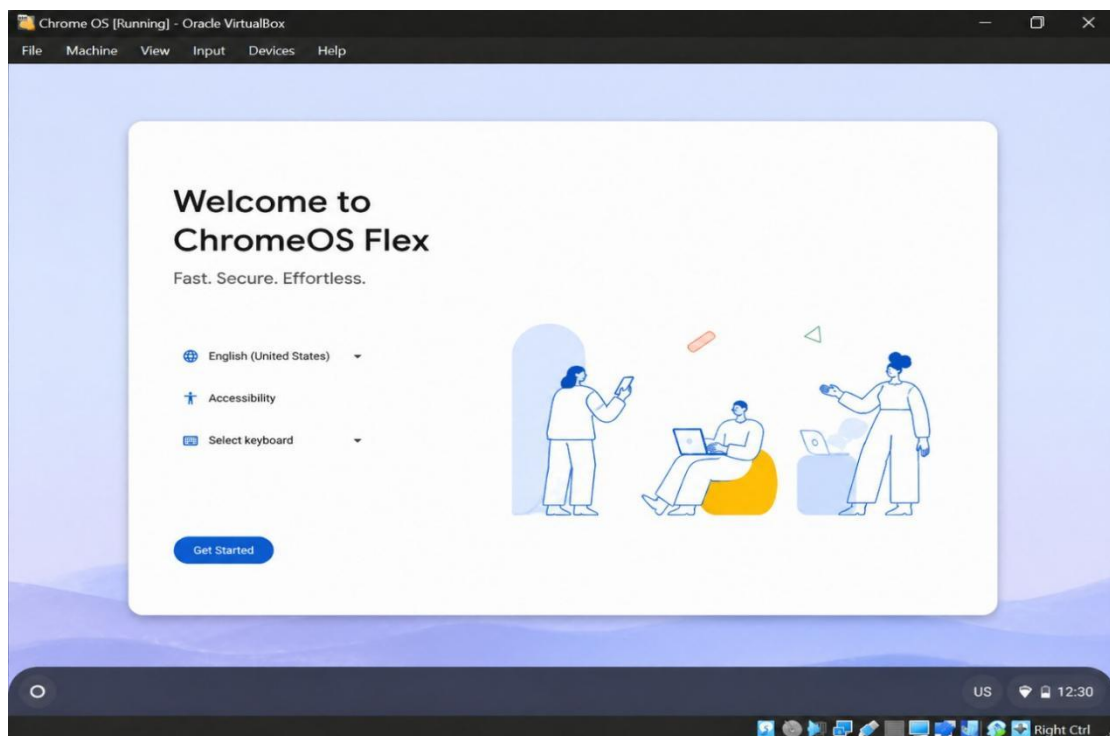


Figure 7: Welcome screen prompting the user to begin the setup process.

Step 8: Network Selection

1. The system scans available Wi-Fi networks.
2. Select the required network.

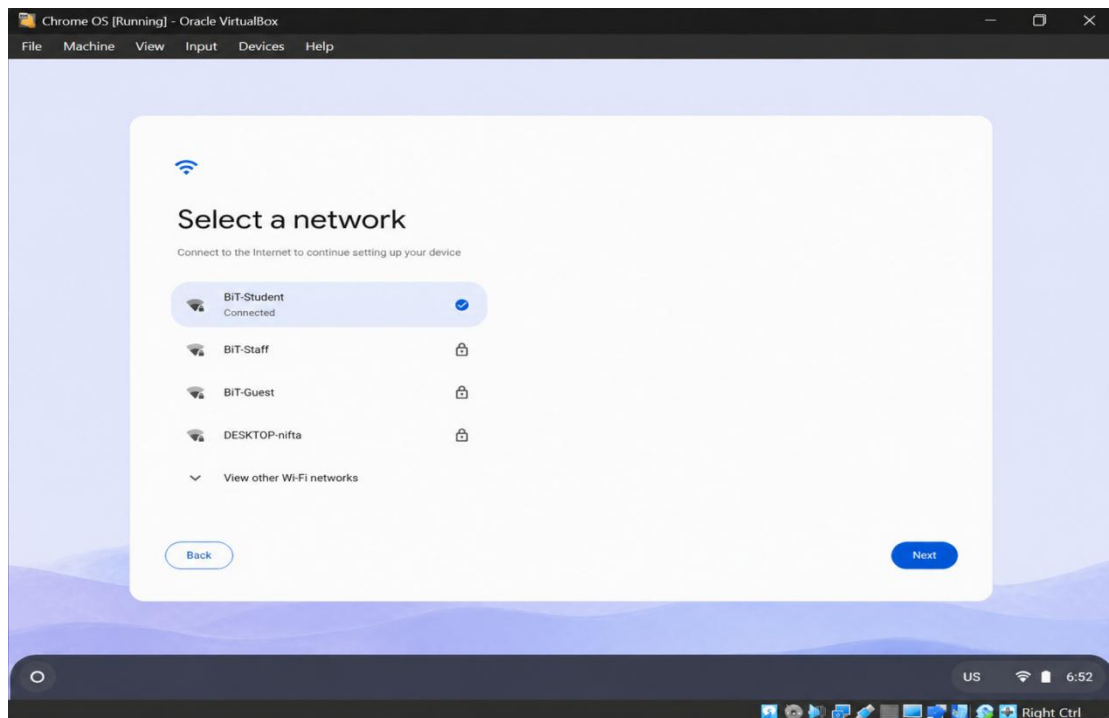


Figure 8: Network selection screen for scanning and connecting to Wi-Fi.

Step 9: Google Account Sign-in

1. Enter Google account credentials.
2. Sign in to continue setup.

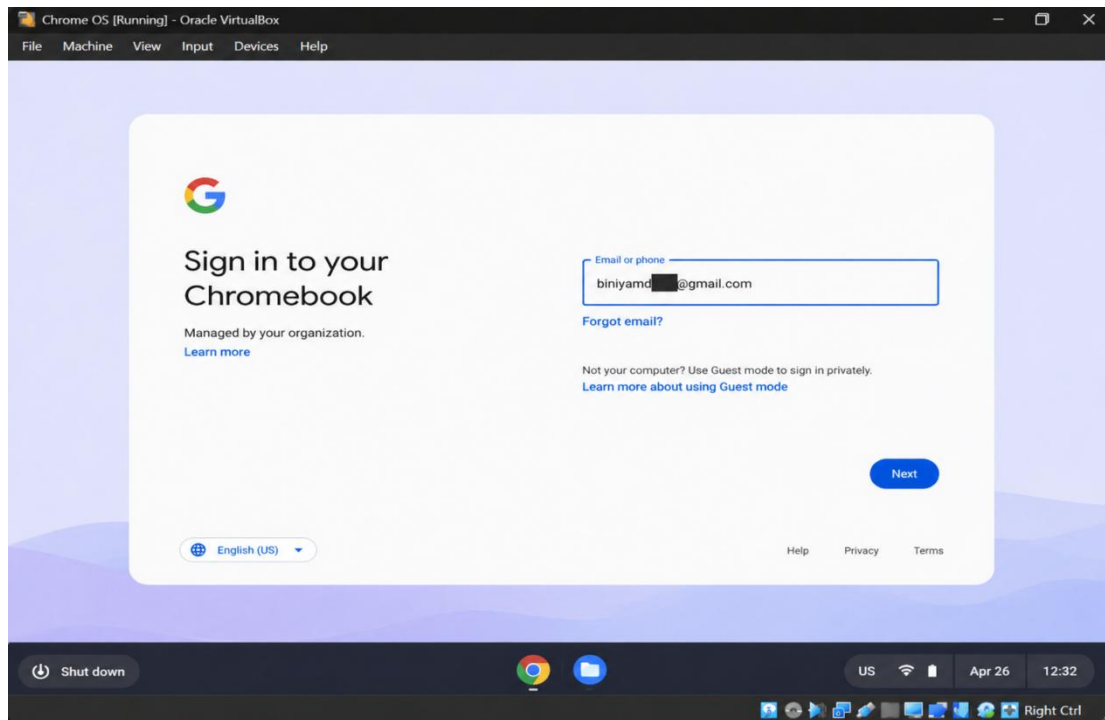


Figure 9: Google account sign-in screen to continue the OS setup.

Step 10: User Setup Screen

1. Configure user profile information.
2. Enter user name and setting

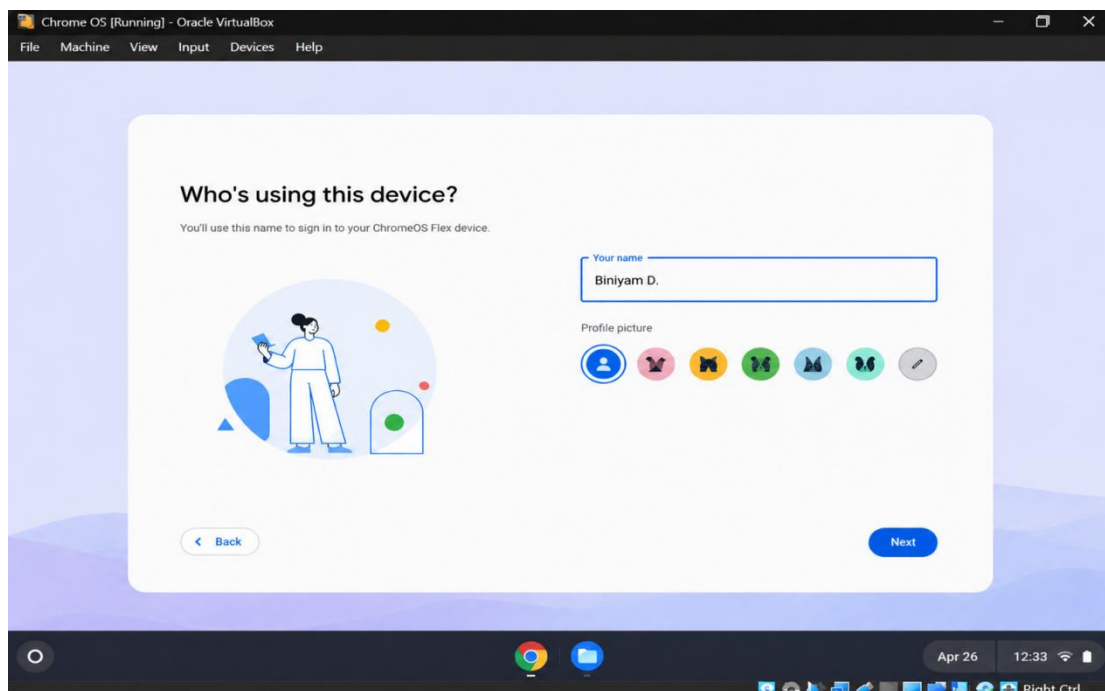


Figure 10: User setup screen for configuring profile information and settings.

Step 11: Final Desktop Screen

1. System completes setup.
2. ChromeOS desktop becomes available.

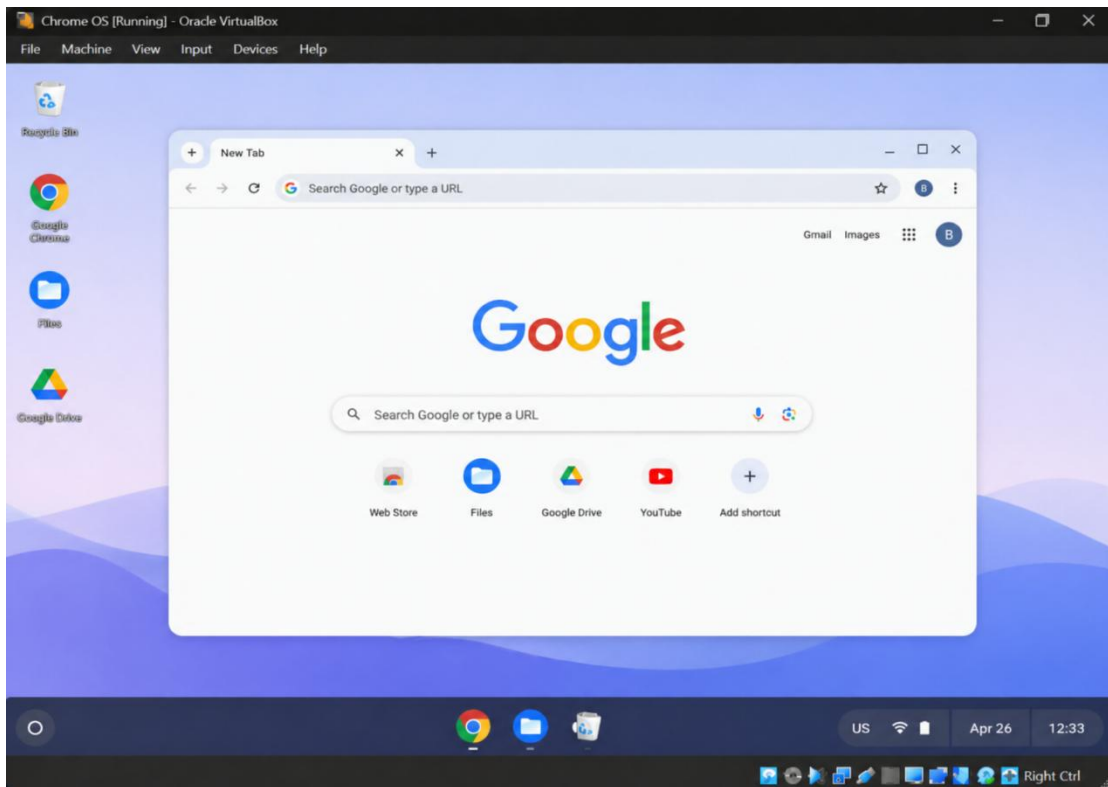


Figure 11: Final desktop screen indicating the completion of the ChromeOS setup.

5. Issues (Problems Faced)

During the installation of ChromeOS Flex on Oracle VirtualBox, several issues were encountered from image preparation to the booting process.

5.1 Error During VirtualBox Installation

During the installation of Oracle VirtualBox, an error message appeared indicating that the **Microsoft Visual C++ 2019 Redistributable Package** was not installed. This dependency is required for VirtualBox to run properly.

As a result, the installation process could not continue until the required package was installed. This issue prevented the successful setup of the virtualization software.

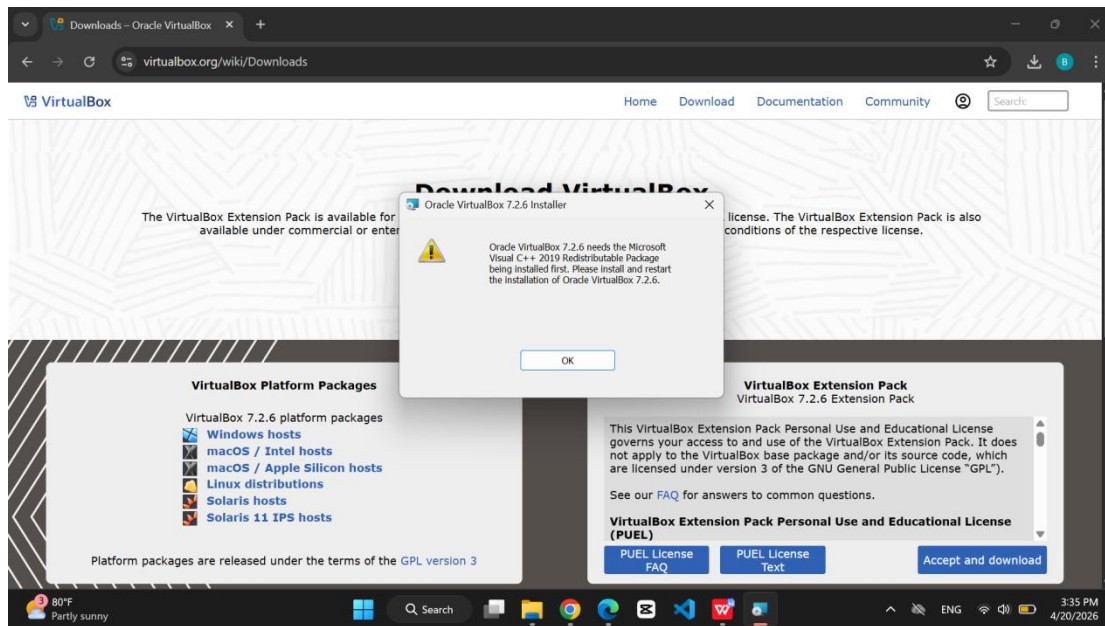
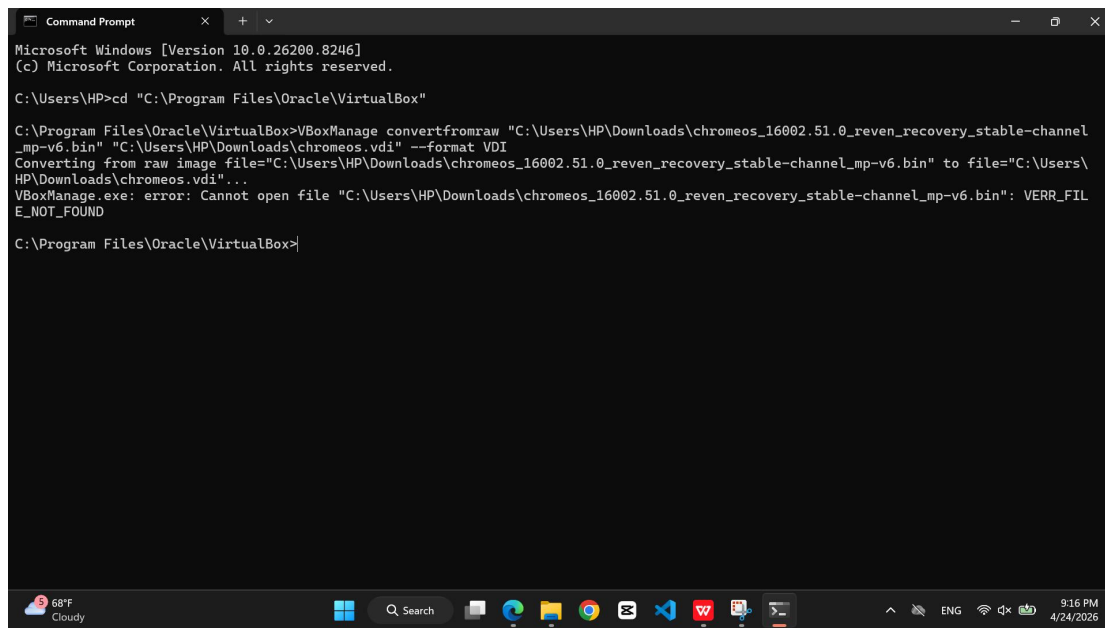


Figure 12: Error message during VirtualBox installation showing missing Microsoft Visual C++ Redistributable dependency

5.2 Error During .BIN to .VDI Conversion

While converting the ChromeOS Flex image file, multiple errors occurred:

- **File not found (VERR_FILE_NOT_FOUND):** caused by incorrect file path and using files inside compressed folders
- **Access denied (VERR_ACCESS_DENIED):** due to system permission restrictions
- **Incorrect file structure:** file was inside nested folders, leading to path confusion



```
Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\HP>cd "C:\Program Files\Oracle\VirtualBox"

C:\Program Files\Oracle\VirtualBox>VBoxManage convertfromraw "C:\Users\HP\Downloads\chromeos_16002.51.0_reven_recovery_stable-channel_mp-v6.bin" "C:\Users\HP\Downloads\chromeos.vdi" --format VDI
Converting from raw image file="C:\Users\HP\Downloads\chromeos_16002.51.0_reven_recovery_stable-channel_mp-v6.bin" to file="C:\Users\HP\Downloads\chromeos.vdi"...
VBoxManage.exe: error: Cannot open file "C:\Users\HP\Downloads\chromeos_16002.51.0_reven_recovery_stable-channel_mp-v6.bin": VERR_FILE_NOT_FOUND

C:\Program Files\Oracle\VirtualBox>
```

Figure 13: Command Prompt showing errors encountered during conversion of ChromeOS Flex .bin image to .vdi format, including file path and access permission issues

6. Solutions

6.1 Solution for VirtualBox Installation Error

The installation issue of Oracle VirtualBox was resolved by installing the required **Microsoft Visual C++ Redistributable Package**. The error message clearly indicated that this dependency was missing.

To fix the problem, the appropriate version (both 32-bit and 64-bit) of the Microsoft Visual C++ 2019 Redistributable was downloaded and installed. After completing the installation, the system was restarted to apply the changes.

Finally, the VirtualBox setup was run again, and the installation completed successfully without any errors.

6.2 .BIN to .VDI Conversion Errors

During the conversion of the ChromeOS Flex image file, several issues were encountered mainly due to incorrect file handling and path errors.

- ✓ To resolve these problems, the image file was first **properly extracted** from the compressed (ZIP) folder, since using files directly from a ZIP archive can lead to access and path errors. Next, the correct file location was ensured by using the

“Copy as path” option instead of typing the path manually, which helped avoid mistakes.

- ✓ Additionally, the Command Prompt was run as an administrator to prevent permission-related errors such as *access denied*. Care was also taken to avoid placing the file inside **nested folders or duplicate file names**, as this can create confusion in the file path and lead to conversion failure.
- ✓ By applying these steps, the .bin file was successfully converted into a .vdi format and used in Oracle VirtualBox.

7. Filesystem Support

Chrome OS Flex utilizes a combination of file systems to ensure efficient internal operations and seamless compatibility with external storage devices. The selection of these file systems is based on performance, reliability, and cross-platform support.

Supported File Systems

1. ext4 (Fourth Extended Filesystem)

The primary file system used internally by Chrome OS Flex is **ext4**, a widely adopted file system in Linux-based operating systems. It is responsible for managing system files, user data, and overall storage operations within the operating system.

2. FAT32 and exFAT

For external storage devices such as USB flash drives and memory cards, Chrome OS Flex supports **FAT32** and **exFAT** file systems. These formats are commonly used across multiple operating systems, making them ideal for data exchange.

Rationale for File System Selection (Why?)

- **ext4** is chosen for internal storage due to its:
 - High reliability and stability
 - Journaling capability, which helps prevent data corruption
 - Efficient performance in handling large volumes of data
- **FAT32 and exFAT** are supported for external devices because they:
 - Provide broad compatibility with various operating systems (e.g., Windows, Linux, macOS)
 - Enable easy transfer of files between different platforms

8. Advantages and Disadvantages

Advantages

- ✓ **Lightweight and Fast Operating System** : Lightweight and Fast Operating System Chrome OS is designed to be minimal, which allows it to boot quickly and run smoothly even on low-end hardware.
- ✓ **High Security (Sandboxing and Verified Boot)** : Chrome OS uses advanced security features such as **sandboxing**, where each application runs in an isolated environment, and **verified boot**, which checks system integrity during startup.
- ✓ **Easy to Use with Cloud Integration** : The interface is simple and integrates well with cloud services like Google Drive for storage and synchronization.
- ✓ **Virtualization Reduces Hardware Dependency** : Running Chrome OS on **Oracle VirtualBox** allows it to operate without needing dedicated hardware. This makes testing, learning, and deployment easier without affecting the host system.

Disadvantages

- ✓ **Limited Offline Functionality**: Many features depend on internet access, so offline usage is restricted.
- ✓ **Dependency on Internet Connectivity**: Core applications require internet, making it less effective without a stable connection.
- ✓ **Limited Support for Traditional Applications**: Traditional desktop software may not run natively and often requires extra setup.
- ✓ **Performance May Be Slower in Virtual Machine**: Performance can decrease due to limited RAM, CPU allocation, and lack of direct hardware access.

9. Conclusion

This project successfully demonstrated the installation and execution of Chrome OS using Oracle VirtualBox with a VDI file. The use of a VDI file simplified the setup process by eliminating the need for manual installation through an ISO file.

Virtualization proved to be an effective and flexible approach for running operating systems without modifying the host system. It allowed safe testing, easy configuration, and efficient resource utilization.

Additionally, the project provided a clear understanding of filesystem support, system requirements, and virtual machine configuration. Overall, it helped in gaining practical knowledge of operating systems and virtualization techniques.

10. Future Outlook / Recommendation

[ChromeOS Flex](#) is expected to continue growing, especially in cloud-based computing environments. As more applications and services move to the cloud, ChromeOS will become more relevant due to its lightweight design and strong integration with online platforms.

The system is highly suitable for students and users who perform basic tasks such as web browsing, document editing, and online learning. Its simplicity and fast performance make it a good choice for educational use.

In addition, ChromeOS Flex is recommended for low-end or older hardware systems, as it requires fewer system resources compared to traditional operating systems. This allows users to extend the life of their devices.

In the future, improvements may focus on enhancing offline capabilities, enabling users to work more effectively without a constant internet connection, and expanding compatibility with more hardware and applications.

11. virtualization in modern operating system

11.1 What is Virtualization?

Virtualization is the process of creating a **virtual (software-based) version** of a physical resource, such as a server, storage device, or operating system.

In a traditional setup, the operating system (OS) interacts directly with the physical hardware. In a virtualized environment, a thin layer of software sits between the hardware and the OS, allowing multiple "Guest" systems to run on a single "Host" machine simultaneously. Each virtual machine (VM) is isolated and behaves as if it has its own dedicated hardware.

11.2 Why Use Virtualization?

Virtualization is a key component of modern operating systems and computing environments due to its numerous benefits:

- ✓ **Efficient Resource Utilization:**
Virtualization allows multiple virtual machines to share a single physical system, thereby maximizing the utilization of hardware resources such as CPU, memory, and storage.
- ✓ **System Isolation and Security:**
Each virtual machine operates independently, ensuring that failures or issues in one environment do not affect others, thus enhancing system security.
- ✓ **Cost Reduction:**
By reducing the need for multiple physical machines, virtualization lowers hardware acquisition, maintenance, and energy costs.
- ✓ **Flexible Testing and Development Environment:**
It provides a controlled and safe environment for testing software, running different operating systems, and conducting research without impacting the host system.

11.3 How Virtualization Works (Working Principle of Virtualization)

Virtualization is implemented through a software layer known as a **hypervisor**, such as Oracle VirtualBox. The hypervisor acts as an intermediary between the physical hardware and virtual machines.

It is responsible for:

- Allocating system resources (CPU, RAM, storage) to each virtual machine
- Managing and monitoring virtual machine operations
- Ensuring isolation between different virtual environments

Each virtual machine created by the hypervisor behaves like an independent computer system, capable of running its own operating system and applications while sharing the underlying physical hardware.

12. Reference

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